

Good morning everyone.

I am Soumyajit, a third-year B.E. Computer Science student at BITS Pilani. During my Practice School-1 at Swecha Foundation, I worked on multiple AI and full-stack projects involving Next.js, FastAPI, Streamlit, machine learning, and generative AI. I will briefly present the projects and focus mainly on my individual contributions.

First, I worked on Adhikar Setu, an AI-powered platform aimed at simplifying policy and incentive access for MSMEs.

The platform helps businesses discover suitable schemes, estimate financial losses, and understand reasons for application rejection.

Tech stack used includes Next.js, FastAPI, Gemini AI, and Vercel.

My major contributions were building the complete frontend and user experience, developing the Loss Calculator module, implementing the AI Rejection Explainer backend, and deploying the application.

Next, I worked on HighOnCode, which is the portfolio website for our team.

It showcases team members, skills, projects, and achievements through an interactive interface.

My contributions included redesigning the AI-generated website, implementing animations, updating the content, improving the overall user experience, and deploying the final portfolio.

I also independently developed the Stock Intelligence Dashboard, a fintech application that simplifies stock analysis and assists users with machine-learning-based predictions.

Tech stack used includes React, TypeScript, FastAPI, Python, Scikit-learn, Plotly, Supabase, Tailwind CSS, and AI APIs.

I designed and implemented the complete application, including stock visualization, technical indicators, prediction modules, watchlists, and AI-assisted analysis. The system helps users understand market trends through interactive charts and intelligent insights.

Another individual project was Quorix, a privacy-preserving PDF question-answering system.

Tech stack includes Next.js, TypeScript, FastAPI, SQLite, PyMuPDF4LLM, and Llama.cpp.

I built the complete project independently. It enables users to upload PDFs, chat with documents locally, explore document hierarchy, and obtain explainable AI responses without relying on vector databases.

I also developed MeowBot AI X, an advanced document intelligence platform.

The technologies used include Streamlit, ChromaDB, Sentence Transformers, Groq API, Plotly, and Python.

I independently implemented the chatbot and analytics system. The platform provides semantic search, keyword extraction, sentiment analysis, readability metrics, and interactive document insights for uploaded files.

Overall, these projects strengthened my understanding of full-stack development, retrieval-augmented generation, machine learning, AI integration, deployment, and building production-ready applications.

Thank you.